Long-term Mortality, Respiratory (Single-Pollutant)

In a single-pollutant model, the coefficient and standard error are based on the warm-season specific hazard ratio of 1.14 (1.10-1.18) per 10 ppb increase in seasonal average of daily 8-hour maximum O_3 concentrations (Turner et al. 2016, Table E5: HBM O_3, 1982-2004, fully adjusted plus ecological covariate).

Long-term Mortality, Respiratory (Multi-Pollutant)

In a multi-pollutant model, the coefficient and standard error are based on the warm-season specific hazard ratio of 1.12 (1.08-1.16) per 10 ppb increase in seasonal average of daily 8-hour maximum O_3 concentrations (Turner et al. 2016, Table E7: HBM O_3, 1982-2004, multipollutant, fully adjusted).

F.7 Sensitivity Analysis – At-Risk Populations

Table F-7 summarizes the ozone health impact functions considered by EPA to be sensitivity analyses that characterize risk experienced by certain subpopulations. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table F-7. Core Health Impact Functions for Ozone Sensitivity Analyses of At-Risk Populations

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
HA, All Respiratory	Cakmak et al.	2006	10 Canadian Cities	0-99		D24HourMean	0.002033	0.000580	Logistic	Female
HA, All Respiratory	Cakmak et al.	2006	10 Canadian Cities	0-99		D8HourMax	0.002033	0.000580	Logistic	Female, 8-hour max from 24-hour mean using adjustment factor of 0.654, resulting in effective beta of 0.001328.
HA, All Respiratory	Cakmak et al.	2006	10 Canadian Cities	0-99		D24HourMean	0.002530	0.000519	Logistic	Male
HA, All Respiratory	Cakmak et al.	2006	10 Canadian Cities	0-99		D8HourMax	0.002530	0.000519	Logistic	Male, 8-hour max from 24-hour mean using adjustment factor of 0.654, resulting in effective beta of 0.001653.
Mortality, Respiratory	Jerrett et al.	2009	Nationwide US, 96 MSAs	30-99		Annual (D1HourMax)	0.003922	0.000972	Log-linear	Female
Mortality, Respiratory	Jerrett et al.	2009	Nationwide US, 96 MSAs	30-99		Annual (D8HourMax)	0.003922	0.000972	Log-linear	Female, 8-hour max from 1-hour max using adjustment factor of 1.13, resulting in effective beta of 0.004432.
Mortality, Respiratory	Jerrett et al.	2009	Nationwide US, 96 MSAs	30-99		Annual (D1HourMax)	0.000995	0.001257	Log-linear	Male
Mortality,	Jerrett et al.	2009	Nationwide	30-99		Annual	0.000995	0.001257	Log-	Male, 8-hour max

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Respiratory			US, 96 MSAs			(D8HourMax)			linear	from 1-hour max using adjustment factor of 1.13, resulting in effective beta of 0.001124.
Mortality, All Cause	Katsouyanni et al.	2009	Nationwide US, 90 cities	0-74		D1HourMax	0.001361	0.000415	Logistic	Age <75
Mortality, All Cause	Katsouyanni et al.	2009	Nationwide US, 90 cities	0-74		D8HourMax	0.001361	0.000415	Logistic	Age <75, 8-hour max from 1-hour max using adjustment factor of 1.13, resulting in effective beta of 0.001542.
Mortality, All Cause	Katsouyanni et al.	2009	Nationwide US, 90 cities	75-99		D1HourMax	0.00121	0.000456	Logistic	Age ≥75
Mortality, All Cause	Katsouyanni et al.	2009	Nationwide US, 90 cities	75-99		D8HourMax	0.00121	0.000456	Logistic	Age ≥75, 8-hour max from 1-hour max using adjustment factor of 1.13, resulting in effective beta of 0.001367.
HA, Lower Respiratory Infection	Lin et al.	2005	Toronto, Canada	0-14		D1HourMax	0.007592	0.005232	Logistic	Female
HA, Lower Respiratory Infection	Lin et al.	2005	Toronto, Canada	0-14		D8HourMax	0.007592	0.005232	Logistic	Female, 8-hour max from 1-hour max using adjustment factor of 1.14, resulting in effective beta of 0.008655.
HA, Lower Respiratory Infection	Lin et al.	2005	Toronto, Canada	0-14		D1HourMax	0.003530	0.004524	Logistic	Male
HA, Lower Respiratory Infection	Lin et al.	2005	Toronto, Canada	0-14		D8HourMax	0.003530	0.004524	Logistic	Male, 8-hour max from 1-hour max using adjustment factor of 1.14, resulting in effective beta of 0.004025.
Emergency Room Visits, Asthma	Mar and Koenig	2009	Seattle, Washington	0-17		D8HourMax	0.010436	0.004358	Log-linear	Age <18
Emergency Room Visits, Asthma	Mar and Koenig	2009	Seattle, Washington	18-99		D8HourMax	0.003922	0.002688	Log-linear	Age ≥18
Mortality, All Cause	Medina-Ramon & Schwartz	2008	Nationwide US, 48 cities	0-64		D8HourMean	-0.000130	0.000102	Logistic	Age <65
Mortality, All Cause	Medina-Ramon & Schwartz	2008	Nationwide US, 48 cities	65-99		D8HourMean	0.000965	0.000235	Logistic	Age ≥65
Mortality, All Cause	Medina-Ramon & Schwartz	2008	Nationwide US, 48 cities	0-99		D8HourMean	0.000936	0.00024	Logistic	Female, Age 0-99

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Mortality, All Cause	Medina-Ramon & Schwartz	2008	Nationwide US, 48 cities	0-99		D8HourMean	0.000359	0.000038	Logistic	Male, Age 0-99
Emergency Room Visits, Asthma	Paulu and Smith	2008	Maine	2-14		D8HourMax	0.010436	0.005027	Logistic	Age 2-14
Emergency Room Visits, Asthma	Paulu and Smith	2008	Maine	15-34		D8HourMax	0.014842	0.003524	Logistic	Age 15-34
Emergency Room Visits, Asthma	Paulu and Smith	2008	Maine	2-99		D8HourMax	0.011333	0.002736	Logistic	Female, Age 2-99
Emergency Room Visits, Asthma	Paulu and Smith	2008	Maine	2-99		D8HourMax	0.010436	0.003222	Logistic	Male, Age 2-99
Emergency Room Visits, Asthma	Villeneuve et al.	2007	Edmonton, Canada	2-4		D8HourMax	0.003237	0.003342	Logistic	Age 2-4
Emergency Room Visits, Asthma	Villeneuve et al.	2007	Edmonton, Canada	5-14		D8HourMax	0.007279	0.002357	Logistic	Age 5-14
Emergency Room Visits, Asthma	Villeneuve et al.	2007	Edmonton, Canada	15-14		D8HourMax	0.005798	0.00179	Logistic	Age 15-14
Emergency Room Visits, Asthma	Villeneuve et al.	2007	Edmonton, Canada	45-64		D8HourMax	0.006296	0.003275	Logistic	Age 45-64
Emergency Room Visits, Asthma	Villeneuve et al.	2007	Edmonton, Canada	65-74		D8HourMax	0.007279	0.005544	Logistic	Age 65-74
Emergency Room Visits, Asthma	Villeneuve et al.	2007	Edmonton, Canada	75-99		D8HourMax	-0.000558	0.006684	Logistic	Age 75-99
Mortality, All Cause	Zanobetti and Schwartz	2008	Nationwide US, 48 cities	0-20		D8HourMean	0.00008	0.000252	Logistic	Age 0-20
Mortality, All Cause	Zanobetti and Schwartz	2008	Nationwide US, 48 cities	21-30		D8HourMean	0.0001	0.000392	Logistic	Age 21-30
Mortality, All Cause	Zanobetti and Schwartz	2008	Nationwide US, 48 cities	31-40		D8HourMean	0.00007	0.000229	Logistic	Age 31-40
Mortality, All Cause	Zanobetti and Schwartz	2008	Nationwide US, 48 cities	41-50		D8HourMean	0.00008	0.000178	Logistic	Age 41-50
Mortality, All Cause	Zanobetti and Schwartz	2008	Nationwide US, 48 cities	51-60		D8HourMean	0.000539	0.000178	Logistic	Age 51-60
Mortality, All Cause	Zanobetti and Schwartz	2008	Nationwide US, 48 cities	61-70		D8HourMean	0.000379	0.000114	Logistic	Age 61-70
Mortality, All Cause	Zanobetti and Schwartz	2008	Nationwide US, 48 cities	71-80		D8HourMean	0.000499	0.000089	Logistic	Age 71-80

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Mortality, All Cause	Zanobetti and Schwartz	2008	Nationwide US, 48 cities	81-99		D8HourMean	0.00029	0.000079	Logistic	Age 81-99

F.7.1 Cakmak et al. (2006)

Cakmak et al. (2006) examined the relationship between daily average O_3 concentrations and hospital admissions for respiratory causes (ICD-9 466, 480-486, 490, 491, 492, 493, 494, and 496) among residents of 10 Canadian cities (Calgary, Edmonton, Halifax, London, Ottawa, Saint John, Toronto, Vancouver, Windsor, and Winnipeg). Data on 215,544 respiratory hospitalizations were obtained from the Canadian Institute for Health Information. Daily 24-hour average O_3 concentrations in all seasons were estimated using the average of data from all available monitors within each city. The authors ran city-specific multi-pollutant Poisson regression models by sex, education level, and income quartile using time lags of 0 to 5 days. Models controlled for day of the week, temperature, humidity, and barometric pressure. Pooled estimates across all 10 cities were calculated by using a random-effects model.

Hospital Admissions, All Respiratory

In single-pollutant models, the coefficient and standard error are estimated from a percentage increase of 3.6% (95% CI: 1.6-5.7%) for females and 4.5% (95% CI: 2.6-6.3%) for males for a 17.4 ppb increase in daily 24-hour average O_3 concentrations (Cakmak et al. 2006, Table 3).

The Health Impact Function was adjusted from the daily 24-hour mean metric to the daily 8-hour max metric using a ratio of 1/1.53 = 0.65359 (inverse of the ratio of 8-hour max to 24-hour mean ozone) (Anderson and Bell 2010, Table 2).

F.7.2 Jerrett et al. (2009)

Jerrett et al. (2009) examined the potential contribution of long-term ozone exposure to the risk of death from cardiopulmonary causes and specifically to death from respiratory causes. Data from the study cohort of the American Cancer Society Cancer Prevention Study II were correlated with air-pollution data from 96 metropolitan statistical areas in the United States. Associations between warm season ozone concentrations and the risk of death were evaluated with the use of standard and multilevel Cox regression models. In single-pollutant models, increased concentrations of either $PM_{2.5}$ or ozone were significantly associated with an increased risk of death from cardiopulmonary causes. In two-pollutant models, $PM_{2.5}$ was associated with the risk of death from cardiovascular causes, whereas ozone was associated with the risk of death from respiratory causes. The estimated relative risk of death from respiratory causes that was associated with an increment in ozone concentration of 10 ppb was 1.040 (95% confidence interval, 1.010 to 1.067). The association of ozone with the risk of death from respiratory causes was insensitive to adjustment for confounders and to the type of statistical model used. The authors concluded that they were not able to